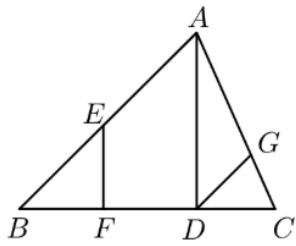


L1- L10 Class & HW

L1

Q10

As shown in the figure below, $\overline{AD} \perp \overline{BC}$, $\overline{EF} \perp \overline{BC}$ at D, F , respectively. $m\angle AEF + m\angle ADG = 180^\circ$, prove: $\overline{AB} \parallel \overline{GD}$



L2

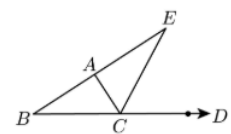
Math Exploration 1 Question 4 Page 20

In triangle ABC , let D and E be on $\overline{AB}$ and $\overline{BC}$, respectively, such that $m\angle BAE = m\angle ACD$. Let F be the intersection of segments $\overline{AE}$ and $\overline{CD}$, and suppose that $\triangle CFE$ is equilateral. What is $m\angle BAC$? (Adapted from 2010 AMC 10A Problem, Question #14)

- A. 60° B. 75° C. 90° D. 105° E. 120°

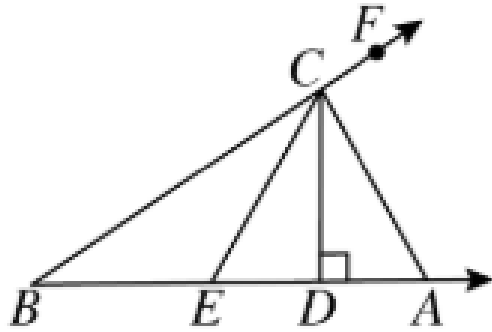
Q2

As shown in the figure below, $\overline{CE}$ is the angle bisector of $\angle ACD$ and $\overline{CE}$ intersects line BA at E , $m\angle B = 32^\circ$, $m\angle E = 26^\circ$, then $m\angle BAC$ is



Q6

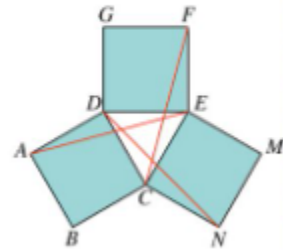
As shown in the figure below, in $\triangle ABC$, $\overline{CD}$ and $\overline{CE}$ are altitude and angle bisector of $\triangle ABC$, $m\angle BAC = 70^\circ$, $m\angle B = 40^\circ$. $m\angle DCE =$ $^\circ$.



L4

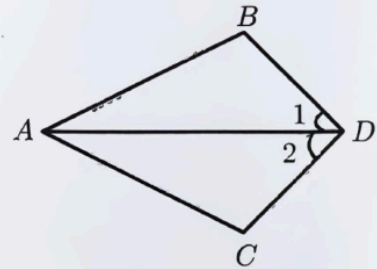
Math Exploration 3

The diagram shows three congruent squares and an equilateral triangle. Prove that $AE = FC = ND$.



Q2

As shown in the figure below, $m\angle 1 = m\angle 2$, which of the following conditions, if added, still cannot deduce $\triangle ABD \cong \triangle ACD$?



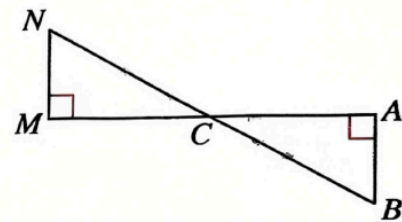
A. $BD = CD$

B. $AB = AC$

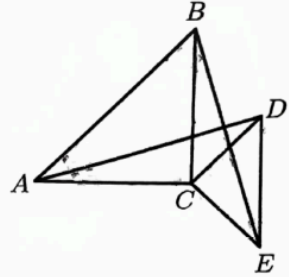
C. $m\angle B = m\angle C$

D. $m\angle BAD = m\angle CAD$

7 Given that $m\angle M = m\angle A$; $NC = BC$, prove that $NM = BA$.



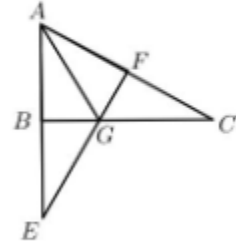
- 9 As shown in the figure below, $\triangle ABC$ and $\triangle ECD$ are isosceles right triangles. $m\angle ACB = m\angle DCE = 90^\circ$, connect $\overline{AD}$ and $\overline{BE}$. Prove that $\triangle ACD \cong \triangle BCE$.



L5

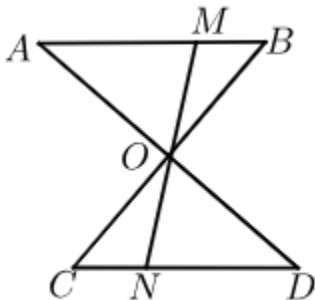
Math Exploration 3

2. In $\triangle ABC$, $m\angle ABC = 90^\circ$. Extend $\overline{AB}$ to E so that $AE = AC$. Create $\overline{EF} \perp \overline{AC}$ at F going through E such that $\overline{EF}$ intersects $\overline{BC}$ at G . Prove that $\overline{AG}$ bisects $\angle BAC$.



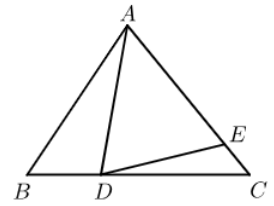
Q3

As shown in the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point O , and $OA = OD$, $OB = OC$. Create $\overline{MN}$ through point O that intersects $\overline{AB}$ at M and $\overline{CD}$ at N . Prove that $OM = ON$.



Q8

It is given that two base angles of an isosceles triangle are equal to each other. As shown in the diagram, then in $\triangle ABC$, $AB = AC$, $m\angle BAD = 30^\circ$, and $AE = AD$. Then $m\angle EDC = \underline{\hspace{1cm}}$.

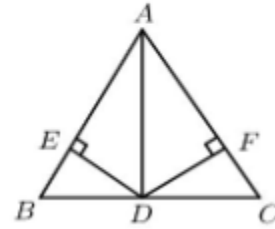


- A) 15
- B) 10
- C) 5
- D) 20

L6

Math Exploration 2

4. As shown in the figure, in $\triangle ABC$, D is the midpoint of $\overline{BC}$. $\overline{DE} \perp \overline{AB}$ and $\overline{DF} \perp \overline{AC}$ at point E and F , respectively. Given that $BE = CF$, prove that $\overline{AD}$ bisects $\angle BAC$.



Math Exploration 3

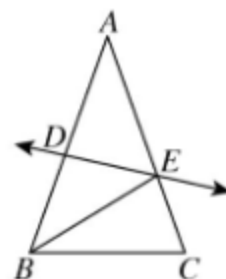
1. As shown in the figure below, in $\triangle ABC$, $m\angle ABC = m\angle C$, $AB = 8$, $\overleftrightarrow{DE}$ is the perpendicular bisector of $\overline{AB}$ and intersects $\overline{AB}$ and $\overline{AC}$ at D , E , respectively. The perimeter of $\triangle BEC$ is 13, $BC = \underline{\hspace{1cm}}$.

A. 5

B. 6

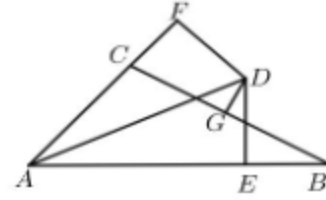
C. 7

D. 8

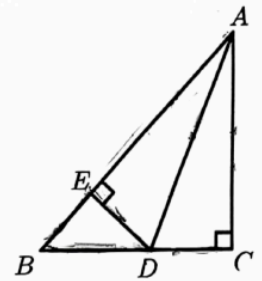


Math Exploration 3

2. As shown in the figure below, $\overline{AD}$ is the angle bisector of $\angle BAC$ and $\overleftrightarrow{DG}$ is perpendicular bisector of $\overline{BC}$. $\overline{AD}$ and $\overleftrightarrow{DG}$ intersect at D . $\overline{DE} \perp \overline{AB}$ at E , $\overline{DF} \perp \overline{AC}$ at F , $AB = 6$, $AC = 3$, then $BE =$ _____. (A little hard, try to connect some points!)



5 As shown in the figure, in $\triangle ABC$, $m\angle C = 90^\circ$, $\overline{AD}$ bisects $\angle BAC$. Draw $\overline{DE} \perp \overline{AB}$ at E through D . If $BC = 9$ and $BE = 3$, then the perimeter of $\triangle BDE$ is _____.



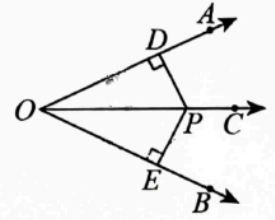
A. 15

B. 12

C. 9

D. 6

- 8 As shown in the figure, $\overrightarrow{OC}$ is inside $\angle AOB$, P is an arbitrary point on the ray OC , $\overline{PD} \perp \overline{OA}$, $\overline{PE} \perp \overline{OB}$. Among the following conditions: ① $m\angle AOC = \frac{1}{2}m\angle AOB$; ② $PD = PE$; ③ $OD = OE$; ④ $m\angle DPO = m\angle EPO$, the number of conditions that can be used to deduce $\overrightarrow{OC}$ is the angle bisector of $\angle AOB$ is _____ .



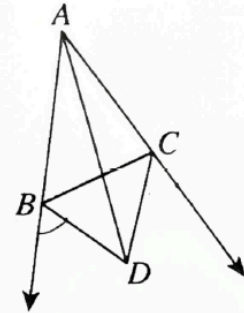
A. 1

B. 2

C. 3

D. 4

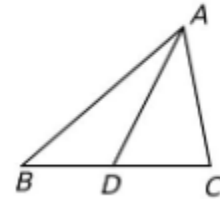
As shown in the figure below, two exterior angle bisectors of $\triangle ABC$ intersect at D .
Prove: $\overline{AD}$ is the angle bisector of $\angle BAC$.



Math Exploration 2

1. As shown in the figure below, in $\triangle ABC$, D is the median of $\overline{BC}$, $AC = 3$, $AD = 4$. Then the length of $\overline{AB}$ cannot be _____.

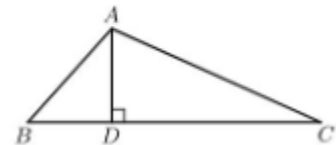
- A. 5 B. 7 C. 8 D. 9



Math Exploration 3

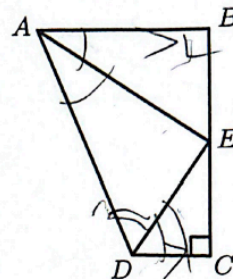
2. As shown in the figure below, in $\triangle ABC$, $m\angle BAC = 108^\circ$, $\overline{AD} \perp \overline{BC}$ at D , $AB + BD = CD$, then $m\angle C$ is _____.

- A. 20° B. 24° C. 30° D. 36°



3

In the quadrilateral $ABCD$ shown in the figure, $\overline{AB} \parallel \overline{CD}$. If the angle bisector of $\angle DAB$, denoted as $\overline{AE}$, intersects $\overline{BC}$ at E , and $\overline{DE}$ bisects $\angle ADC$. Determine whether the following conclusion is true or false.



1. $CD + AB = BC$

2. E is the midpoint of $\overline{BC}$

3. $m\angle AED = 90^\circ$ ✓ CP

4. the area of triangle $ADE = \frac{1}{2}$ the area of quadrilateral $ABCD$

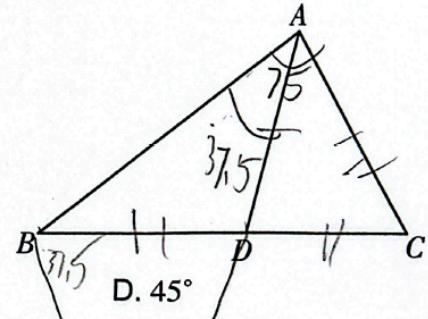
- 4 In triangle ABC , $\overline{AD}$ is the angle bisector of $\angle BAC$, and $AB = AC + CD$. If $m\angle BAC = 75^\circ$, then the measure of $\angle B$ is _____.

A. 25°

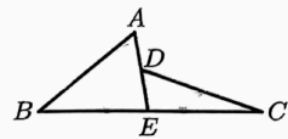
B. 35°

C. 37.5°

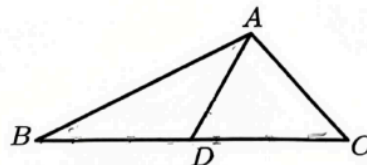
D. 45°



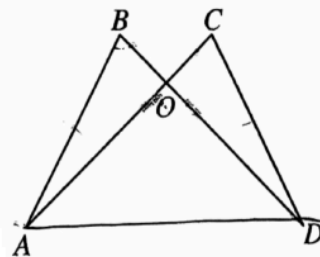
- 5 As shown below, $\angle A = 60^\circ$, point E is a midpoint on line BC . D is a point on line AE . $AB = CD = 3$. Find the value of $AE + DE =$ _____.



- 7 In $\triangle ABC$, $BD = CD$, $\angle DAC = \angle ABC + \angle ACB$. Given that $AD = 3$, find the length of $AB = \underline{\hspace{2cm}}$.



- 10 As shown in the figure below, $AC = DB$, $m\angle B = m\angle C$, prove: $AB = CD$.



Math Exploration 2

2. How many right triangles have integer leg lengths a and b and a hypotenuse of length $b + 1$, where $b < 100$? (2008 AMC 10B Problem, Question #15)

- A. 6 B. 7 C. 8 D. 9 E. 10

Math Exploration 3

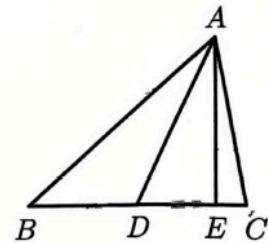
3. Point B is due east of point A . Point C is due north of point B . The distance between points A and C is $10\sqrt{2}$, and $\angle BAC = 45^\circ$. Point D is 20 meters due north of point C . The distance AD is between which two integers? (2012 AMC 10B Problem, Question #12)

- A. 30 and 31 B. 31 and 32 C. 32 and 33 D. 33 and 34 E. 34 and 35

- 2 A rectangular parking lot has a diagonal of 25 meters and an area of 168 square meters. In meters, what is the perimeter of the parking lot?(2011 AMC 10B Problem, Question #14)

A. 52 B. 58 C. 62 D. 68 E. 70

- 8 As shown in the figure, in $\triangle ABC$, $\overline{AD}$ is the median from A to $\overline{BC}$, and $\overline{AE} \perp \overline{BC}$ at point E . If $AB = 12$, $BC = 10$, and $AC = 8$, then DE is _____.



Math Exploration 2

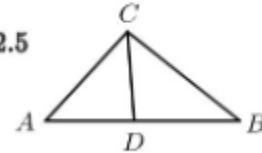
1. In $\triangle ABC$, $\overline{CD}$ is the angle bisector of $\angle ACB$, and $CA = 3$, $CB = 5$, $AB = 6$. The length of $\overline{AD}$ is _____.

A. 2

B. 3

C. $\frac{9}{4}$

D. 2.5



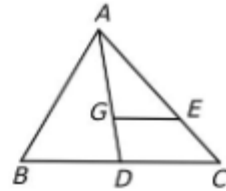
Math Exploration 2

2. Trapezoid $ABCD$ has $\overline{AD} \parallel \overline{BC}$, $BD = 1$, $m\angle DBA = 23^\circ$, and $m\angle BDC = 46^\circ$. The ratio $BC : AD$ is $9 : 5$. What is CD ? (2009 AMC 12B Problem, Question #16)

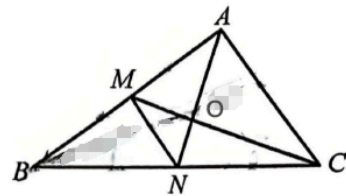
A. $\frac{7}{9}$ B. $\frac{4}{5}$ C. $\frac{13}{15}$ D. $\frac{8}{9}$ E. $\frac{14}{15}$

Math Exploration 3

1. As shown in the figure below, point G is the centroid of triangle ABC , and $\overline{GE}$ is parallel to $\overline{BC}$. If $BC = 12$, then $GE = \underline{\hspace{2cm}}$.

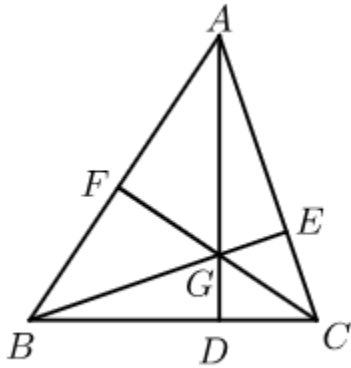


5. As shown in the figure below, O is the centroid of triangle ABC . $\overline{AN}$ and $\overline{CM}$ intersect at point O . The area of triangle MON is 1. Find the area of triangle ABC .



Q2

$\overline{AD} \perp \overline{BC}$ at point D , $\overline{BE} \perp \overline{AC}$ at point E , $\overline{CF} \perp \overline{AB}$ at point F , and $\overline{AD}$, $\overline{BE}$, $\overline{CF}$ intersect at point G .

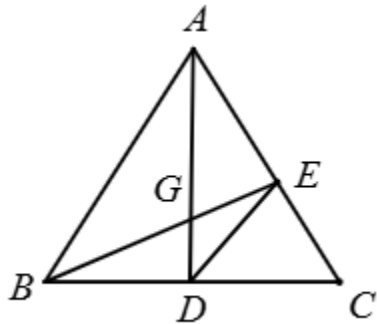


(1) Point G is the orthocenter of triangle _____ .

(2) Point A is the orthocenter of triangle _____ .

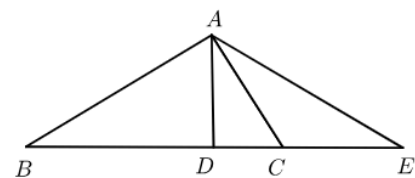
Q3

In the figure, it is known that $\overline{AD}$ is the median of triangle ABC , G is the centroid of triangle ABC . Connect $\overline{BG}$ and extend it to intersect side $\overline{AC}$ at point E . Connect $\overline{DE}$. Then, the ratio of the area of triangle ABC to the area of triangle GED is



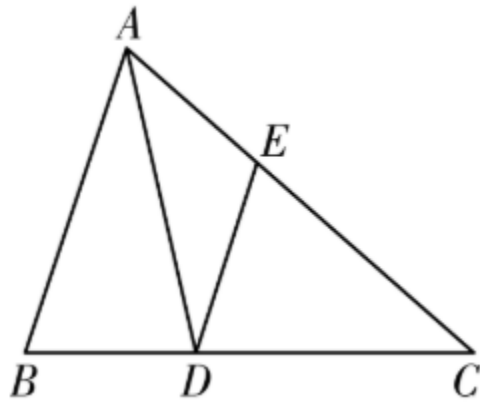
Q4

In the figure, in right triangle ABC , $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at D , construct $m\angle EAC = m\angle B$, and the extension of $\overline{AE}$ intersects with $\overline{BC}$ at E . If $AD = 3$ and $DE = 4$, then $DC =$, $AC =$.



Q8

Given the diagram, it is known that $\overline{AD}$ is the angle bisector of $\triangle ABC$, and $\overline{DE} \parallel \overline{AB}$. If $\frac{AE}{EC} = \frac{2}{3}$, find the value of $\frac{AC}{AB}$.



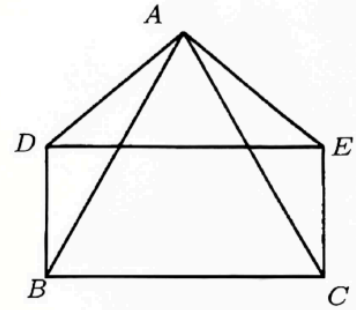
Q10

In triangle ABC , $m\angle A = (x^2 + 6x + 12)^\circ$, $m\angle B = (3x + 8)^\circ$, $m\angle C = (18x)^\circ$, find the longest side of $\triangle ABC$.

L10

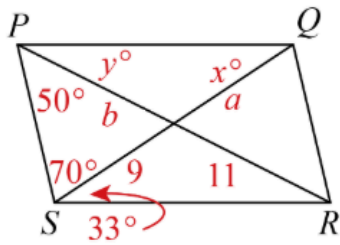
- As shown in the figure, it is given that $AB = AC$, $AD = AE$, $DE = BC$, and $m\angle BAD = m\angle CAE$.

Prove that quadrilateral $BCED$ is a rectangle.



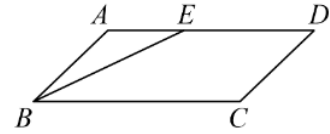
Q1

As shown in the figure below, $PQRS$ is a parallelogram. Find all the unknown angles and sides in it.



Q2

As shown in the figure, in parallelogram $ABCD$, E is the midpoint of side $\overline{AD}$. If $m\angle ABE = m\angle EBC$, $AB = 2$, then the perimeter of $ABCD$ is _____.



8 Which of the following conditions can be used to prove that parallelogram $ABCD$ is a rectangle?

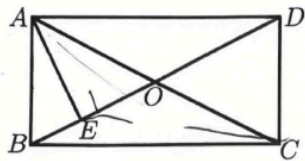
A. $CD = AB$

B. $m\angle A = m\angle C$

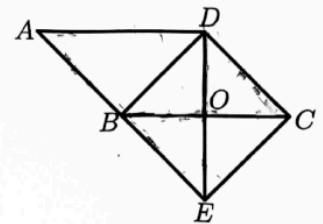
C. $AD = AB$

D. $AC = BD$

- 9 As shown, in rectangle $ABCD$, diagonals $\overline{AC}$ and $\overline{BD}$ intersect at O . $\overline{AE} \perp \overline{BD}$ at point E . $m\angle DAE : m\angle BAE = 3 : 1$. Then $m\angle EAC = \underline{\hspace{1cm}}^\circ$.



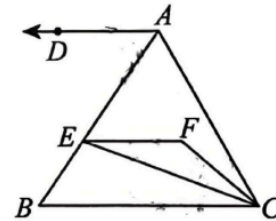
- 10 As shown in the figure below, in parallelogram $ABCD$, extend side $\overline{AB}$ to point E such that $BE = AB$. Draw $\overline{DE}$, $\overline{BD}$, and $\overline{EC}$. $\overline{DE}$ intersects with $\overline{BC}$ at point O . $m\angle BOD = 2m\angle A$. Prove that quadrilateral $BECD$ is a rectangle.



Extention

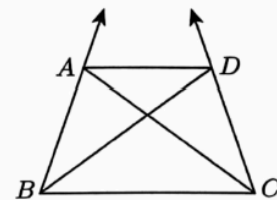
L1

- 10 As shown in the figure below, $\overline{EF} \parallel \overline{AD}$, $\overline{AD} \parallel \overline{BC}$, $\overline{CE}$ bisects $\angle BCF$, $m\angle DAC = 3\angle BCF$, $m\angle ACF = 20^\circ$. Find $m\angle FEC$.



L1 E

- 7 As shown in the figure below, given four conditions: ① $\overline{AC} = \overline{BD}$, ② $m\angle DAC = m\angle BCA$, ③ $m\angle ABD = m\angle CDB$, and ④ $m\angle ADB = m\angle CBD$, the conditions which can be used to prove $\overline{AD} \parallel \overline{BC}$ are _____.



A ①②

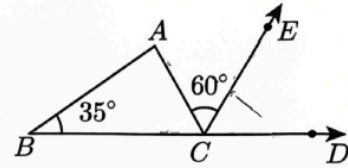
B. ③④

C ②④

D. ①③④

L2 E

- 7 As shown in the figure below, $\overrightarrow{CE}$ is the bisector of $\angle ACD$. If $m\angle B = 35^\circ$, $m\angle ACE = 60^\circ$, then $m\angle A$ is _____.



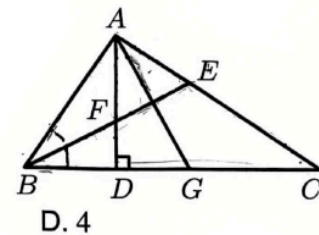
A. 35°

B. 95°

C. 85°

D. 75°

- 10 As shown in the figure, in $\triangle ABC$, $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$, the bisector of $\angle ABC$, $\overline{BE}$, intersects $\overline{AD}$ at point F , and $\overline{AG}$ bisects $\angle DAC$. Given the following statements: ① $m\angle BAD = m\angle C$; ② $m\angle AEF = m\angle AFE$; ③ $m\angle EBC = m\angle C$; ④ $\overline{AG} \perp \overline{EF}$. The number of correct statement(s) is _____.



A. 1

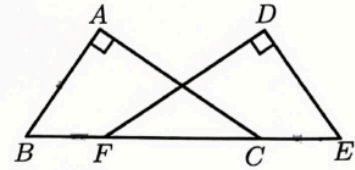
B. 2

C. 3

D. 4

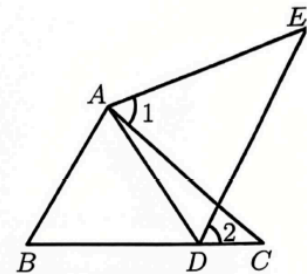
L3

- 8 As shown in the figure below, $m\angle A = m\angle D = 90^\circ$, $AB = DE$, $BF = EC$. Prove: $\text{Rt}\triangle ABC \cong \text{Rt}\triangle DEF$.



L3 E

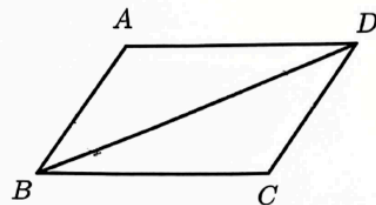
- 9 As shown in the figure below, $m\angle 1 = m\angle 2$, in order to prove $\triangle ABC \cong \triangle ADE$, we can add condition _____.



- A. $AB = AD$, $AC = AE$
C. $AB = DE$, $BC = AE$

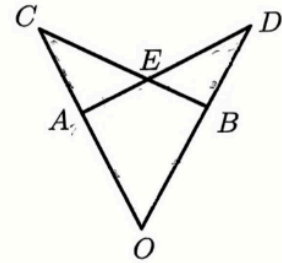
- B. $AB = AD$, $BC = DE$
D. $AC = AE$, $BC = DE$

- 10 As shown in the figure below, $AB = CD$, $\overline{AB} \parallel \overline{CD}$. Prove that $AD = BC$.



L4

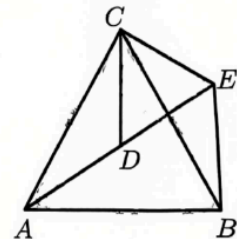
- 6 As shown in the diagram, $OA = OB$, $OC = OD$, $m\angle O = 60^\circ$, $m\angle C = 25^\circ$. The measure of $\angle BED$ is _____.



.. 70°
 C. 65°

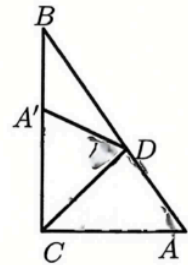
B. 85°
 D. None of the above

- 13 As shown in the figure below, $\triangle ACB$ and $\triangle DCE$ are equilateral triangles. D and E are on the same line. Connect $\overline{BE}$. $m\angle AEB$ is _____.



L4 E

- 5 As shown in the figure, in $\text{Rt}\triangle ABC$, $m\angle ACB = 90^\circ$, $m\angle A = 55^\circ$. Fold the triangle such that A is on $\overline{CB}$ at the position marked as A' . The trace is $\overline{CD}$, then $m\angle A'DB = \underline{\hspace{1cm}}$.



A. 40°

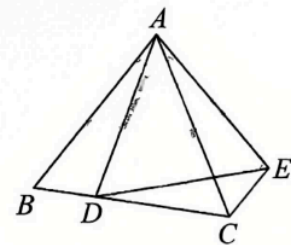
B. 30°

C. 20°

D. 10°

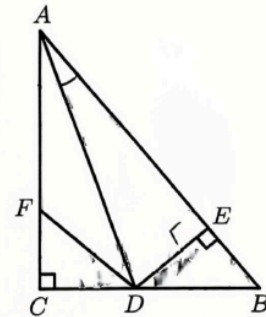
- 6 As shown in the figure below, point D is on $\overline{BC}$, $\triangle ABC$ and $\triangle ADE$ are equilateral triangles.

Prove: $m\angle ACE = m\angle BAC$.



L5

- 2 As shown in the figure below, in $\text{Rt}\triangle ABC$, $m\angle ACB = 90^\circ$, $m\angle B = 50^\circ$, D and F are points on $\overline{BC}$ and $\overline{AC}$, respectively. $\overline{DE} \perp \overline{AB}$ at E , $CF = BE$, $DF = DB$, then $m\angle ADE$ is _____.



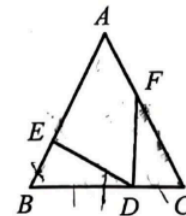
A. 40°

B. 50°

C. 70°

D. 71°

- 3 As shown in the figure below, in $\triangle ABC$, $AB = AC$, $BE = CD$, $BD = CF$, then $m\angle EDF =$ _____.



A. $2m\angle A$

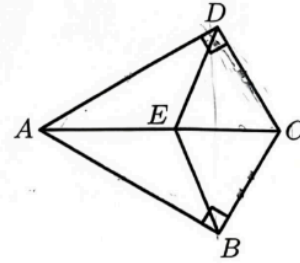
B. $90^\circ - 2m\angle A$

C. $90^\circ - m\angle A$

D. $90^\circ - \frac{1}{2}m\angle A$

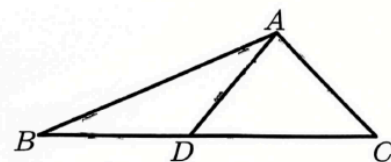
- 9 As shown in the figure below, given that $m\angle ABC = m\angle ADC = 90^\circ$, E is a point on $\overline{AC}$, $AB = AD$.

Prove: $EB = ED$.

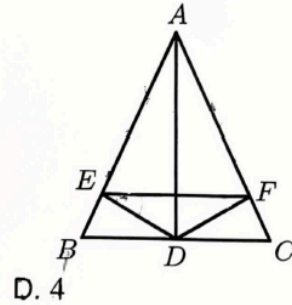


L5 E

- 2 As shown in the figure, in $\triangle ABC$, D is on $\overline{BC}$ and $AC = AD = DB$. If $m\angle BAC = 102^\circ$, then $m\angle ADC = \underline{\hspace{1cm}}^\circ$.



- 3 As shown in the figure below, in $\triangle ABC$, $AB = AC$, $\overline{AD}$ bisects $\angle BAC$, $\overline{DE} \perp \overline{AB}$ at E and $\overline{DF} \perp \overline{AC}$ at F . Among the four following conditions: (1) $m\angle DEF = m\angle DFE$; (2) $AE = AF$; (3) $\overline{AD}$ bisects $\angle EDF$; (4) $\overline{EF}$ perpendicular bisects $\overline{AD}$, how many them are correct?



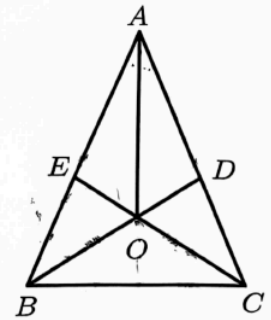
A. 1

B. 2

C. 3

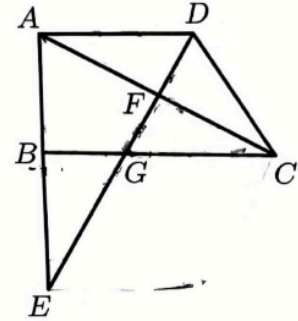
D. 4

- 6 As shown in the figure below, in $\triangle ABC$, $AB = AC$, $\overline{BD}$ and $\overline{CE}$ are angle bisectors of $\angle ABC$ and $\angle ACB$, respectively and they intersect at O . Connect $\overline{OA}$.

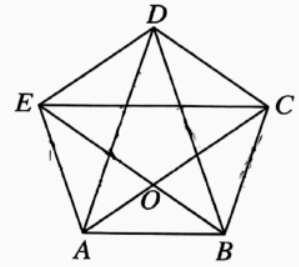


(2) Prove that line OA is perpendicular to $\overline{BC}$.

- 8 As shown in the figure, in quadrilateral $ABCD$, $\overline{AD} \parallel \overline{BC}$, $m\angle ABC = 90^\circ$, $\overline{DE} \perp \overline{AC}$ at point F , intersecting $\overline{BC}$ at point G and $\overline{AE}$ at E . $AE = AC$. Prove that $BG = FG$. (Hint: Connect $\overline{AG}$)



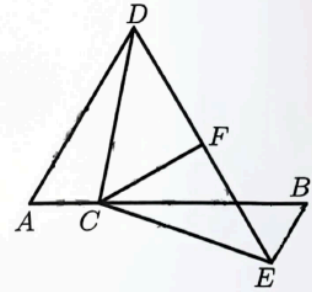
- 10 As shown in the figure below, $\overline{AE} \parallel \overline{BD}$, $\overline{BC} \parallel \overline{AD}$, $AE = BC$ and $AD = BD$, prove that:
- (1) $AC = BE$



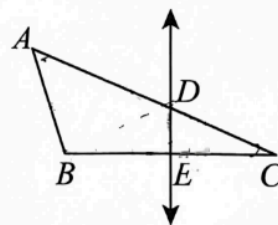
- (2) $\overline{EC} \parallel \overline{AB}$

L6

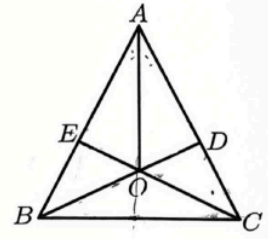
- 3 As shown in the figure below, point C is on segment AB , $\overline{AD} \parallel \overline{EB}$, $AC = BE$, $AD = BC$, $\overline{CF}$ bisects $\angle DCE$. Prove: $\overline{CF} \perp \overline{DE}$.



- 6 As shown in the figure below, in $\triangle ABC$, perpendicular bisector of $\overline{BC}$ intersects $\overline{AC}$ and $\overline{BC}$ at D and E , respectively. If $AB = CD$, $m\angle C = 20^\circ$, then $m\angle A = \underline{\hspace{2cm}}$.

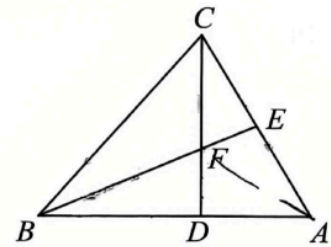


- 7 As shown in the figure below, in $\triangle ABC$, $\overline{BD} \perp \overline{AC}$ at D , $\overline{CE} \perp \overline{AB}$ at E . $BD = CE$ and they intersect at O . Connect $\overline{AO}$. Prove: $\overleftrightarrow{AO}$ is the perpendicular bisector of BC .



- 8 As shown in the figure below, in $\triangle CBD$, $CD = BD$, $\overline{CD} \perp \overline{BD}$, $\overline{BE}$ bisects $\angle CBA$ and intersects $\overline{CD}$ at F . $\overline{CE} \perp \overline{BE}$ at E , $\overline{CE}$ intersects $\overline{BD}$ at A .

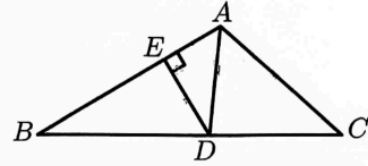
(1) Prove: $BF = AC$.



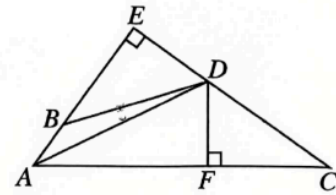
(2) $\overleftrightarrow{BE}$ is the perpendicular bisector of $\overline{AC}$.

L6 E

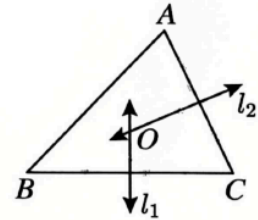
- 4 As shown in the figure, $\overline{AD}$ is the bisector of $\angle BAC$ in $\triangle ABC$. If $\overline{DE} \perp \overline{AB}$ at point E , $DE = 2$ and $AC = 3$, then the area of $\triangle ADC$ is _____.



- 6 As shown in the figure, $BE = CF$, $\overline{DE} \perp \overline{AB}$ and intersects line AB at point E . $\overline{DF} \perp \overline{AC}$ at point F and $DB = DC$. Prove that $\overline{AD}$ is the bisector of $\angle EAC$.

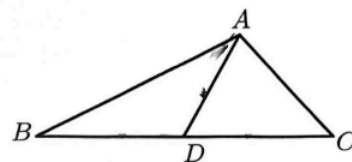


- 10 As shown in the figure below, l_1 is the perpendicular bisector of $\overline{BC}$ and l_2 is the perpendicular bisector of $\overline{AC}$. l_1 and l_2 intersect at O . Prove: Point O is on the perpendicular bisector of $\overline{AB}$.

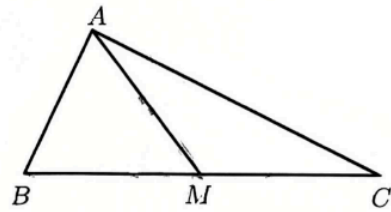


L7

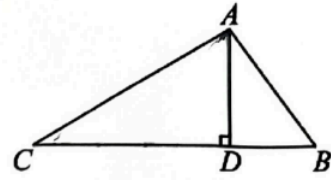
- 5 In $\triangle ABC$, $BD = CD$, $\angle DAC = \angle ABC + \angle ACB$. Given that $AD = 3$, find the length of $AB = \underline{\hspace{2cm}}$.



- 6 As shown in the figure, in $\triangle ABC$, $\overline{AM}$ is the median. Prove that $AM < \frac{1}{2}(AB + AC)$.

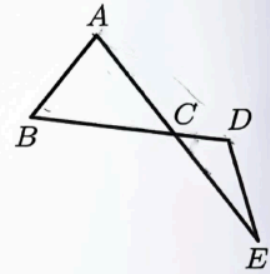


- 9 In $\triangle ABC$, AD is perpendicular BC at point D . Given that $CD = AB + BD$. Prove that $\angle B = 2\angle C$.

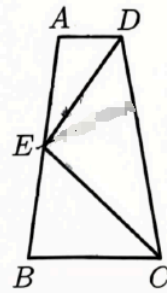


L7 E

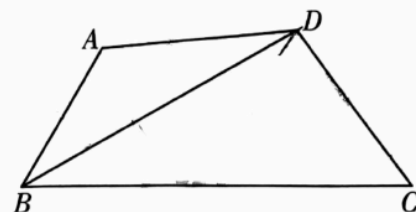
27 As shown in the figure below, $m\angle B + m\angle CDE = 180^\circ$, $AC = CE$. Prove: $AB = DE$.



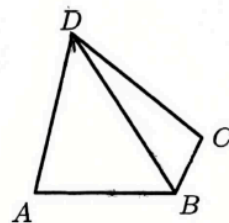
- 4 As shown in the figure, in trapezoid $ABCD$, $\overline{AD} \parallel \overline{BC}$, E is the midpoint of $\overline{AB}$, $\overline{DE} \perp \overline{CE}$, $AD = 3$, and $CD = 9$. Then $BC = \underline{\hspace{2cm}}$.



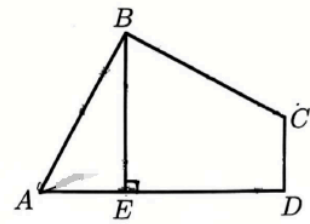
- 6 As shown in the figure, given that in quadrilateral $ABCD$, $BC > AB$, $AD = CD$, $\overline{BD}$ bisects $\angle ABC$, prove that $m\angle A + m\angle C = 180^\circ$.



- 7 As shown in the figure, in $ABCD$, $\overline{BD}$ bisects $\angle ABC$ and $m\angle A + m\angle C = 180^\circ$. If $AB > BC$, prove that $AD = DC$.

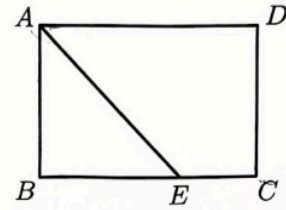


- 9 In the figure, in quadrilateral $ABCD$, $AB = BC$, $m\angle ABC = m\angle CDA = 90^\circ$, $\overline{BE} \perp \overline{AD}$ at E , and the area of quadrilateral $ABCD$ is 10. The length of $\overline{BE}$ is _____.



L8

- 5 In rectangle $ABCD$, $AB = 3$, $BC = 4$, and E is a point on $\overline{BC}$ which is not at point B or point C . If AE is an integer, $AE = \underline{\hspace{1cm}}$.



A. 3

B. 4

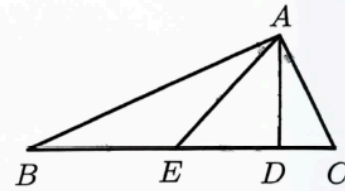
C. 5

D. 6

- 8 In triangle ABC , $AB = AC$, $AC = 2$, and $m\angle BAC = 30^\circ$. Find the length of $\overline{BC}$.

L8 E

- 7 As shown in the figure, in $\triangle ABC$, $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at point D . E is the midpoint of $\overline{BC}$, and $AD = ED = 3$. The length of $\overline{BC}$ is _____.

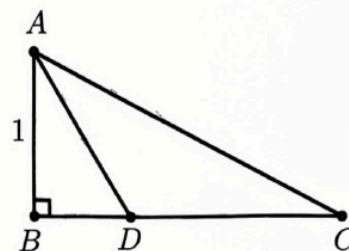


- A. $6\sqrt{2}$ B. $4\sqrt{3}$ C. $3\sqrt{2}$ D. $3\sqrt{2} + 2\sqrt{3}$

- 2 Right triangle ABC has leg lengths $AB = 20$ and $BC = 21$. Including $\overline{AB}$ and $\overline{BC}$, how many line segments with integer length can be drawn from vertex B to a point on hypotenuse $\overline{AC}$? (2018 AMC 10A Problem, Question #16)

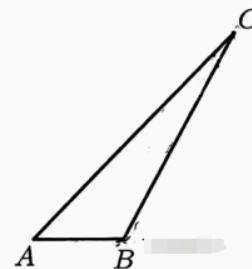
- A. 5 B. 8 C. 12 D. 13 E. 15

- 4 Triangle ABC has a right angle at B , $AB = 1$, and $BC = 2$. The bisector of $\angle BAC$ meets $\overline{BC}$ at D . What is BD ? (2009 AMC 10B Problem, Question #20)



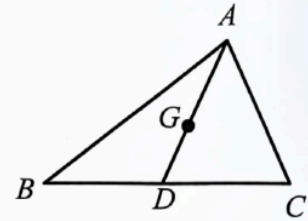
- A. $\frac{\sqrt{3}-1}{2}$ B. $\frac{\sqrt{5}-1}{2}$ C. $\frac{\sqrt{5}+1}{2}$ D. $\frac{\sqrt{6}+\sqrt{2}}{2}$ E. $2\sqrt{3}-1$

- 10 As shown in the figure, in $\triangle ABC$, $m\angle A = 45^\circ$, $m\angle C = 15^\circ$, and $AB = 10$. Calculate the area of $\triangle ABC$.



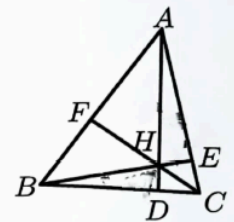
L9

2. As shown in the figure below, in $\triangle ABC$, $\overline{AD}$ is the median from A to $\overline{BC}$ and G is the centroid. If $AG = 6$, then the length of $\overline{DG}$ is _____.



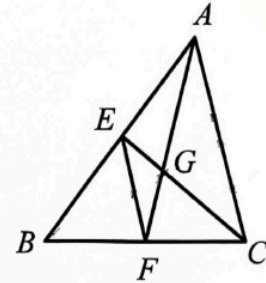
- A. 1 B. 2 C. 3 D. 6

1. In the figure, H is the orthocenter (the intersection point of the three altitudes) of acute triangle ABC . If $AH = BC$, then the degree measure of angle BAC is _____.



L9 E

- 2 In the figure, point G is the centroid of triangle $\triangle ABC$. Connect $\overline{CG}$ and $\overline{AG}$, and extend them to intersect $\overline{AB}$ and $\overline{BC}$ at points E and F . Connect $\overline{EF}$. If $AC = 3.4$, the length of $\overline{EF}$ is _____.

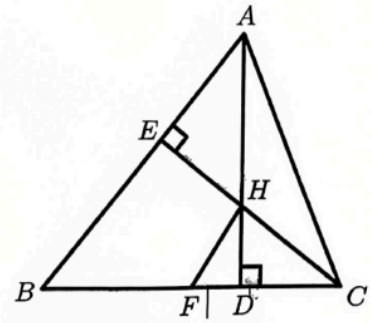


- A. 1.7 B. 1.8 C. 2.2 D. 3.4

- 3 The lengths of the three sides of the triangle are 10, 10, and $10\sqrt{3}$. Find the distance from the incenter of a triangle to its longest side.

- 5 In the acute triangle ABC , let $\overline{AD}$ and $\overline{CE}$ be the altitudes, and let H be the orthocenter. Given that $AD = BC$ and F is the midpoint of $\overline{BC}$, prove that:

$$FH + DH = \frac{1}{2}BC.$$



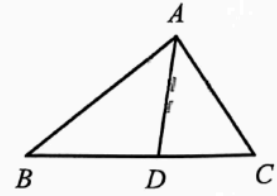
6 In the isosceles triangle $\triangle ABC$, $AB = AC = 6$, $BC = 6\sqrt{3}$, and I is the incenter of $\triangle ABC$.

(1) Find $m\angle BIC$.

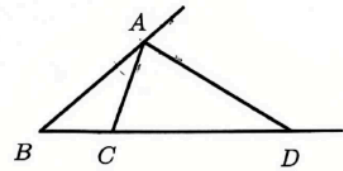
(2) Find AI .

8 Use parallel line to prove the following:

(1) As shown in the figure below, $\overline{AD}$ is the angle bisector of $\triangle ABC$, prove that $\frac{AB}{AC} = \frac{BD}{CD}$.



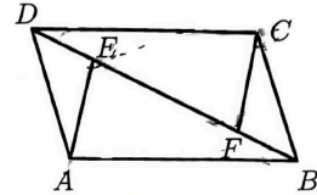
(2) In the figure, the external angle bisector of $\angle BAC$ intersects the extension of side $\overline{BC}$ at point D. Prove that $\frac{AB}{AC} = \frac{BD}{CD}$.



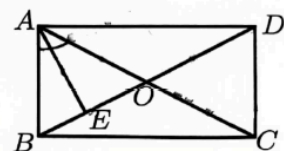
L10



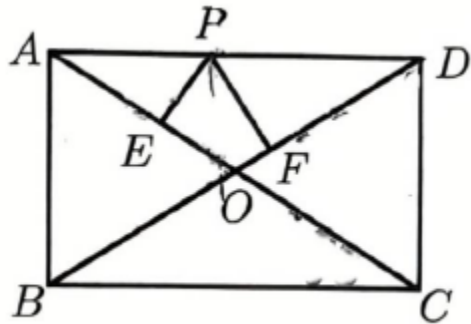
As shown in the figure, in parallelogram $ABCD$, points E and F are on $\overline{BD}$, and $DF = BE$. Hypothesize the positional and quantitative relationships between $\overline{AE}$ and $\overline{CF}$ and prove them.



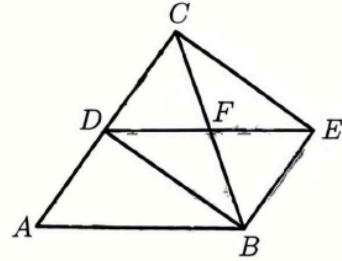
- 8 As shown in the figure, in rectangle $ABCD$, diagonals $\overline{AC}$ and $\overline{BD}$ intersect at point O . Draw $\overline{AE} \perp \overline{BD}$ through A such that the perpendicular foot is located at E . If $m\angle OCD = 56^\circ$, then $m\angle EAO = \underline{\hspace{1cm}}^\circ$.



- 9 As shown in the figure, in rectangle $ABCD$, $AB = 3$, $AD = 4$, and P is a moving point on $\overline{AD}$. Given that $\overline{PF} \perp \overline{BD}$ at point F and that $\overline{PE} \perp \overline{AC}$ at point E , then $PE + PF$ is _____.



- 10 As shown in the figure, in $\triangle ABC$, $AB = BC$, $\overline{BD}$ bisects $\angle ABC$. Draw a parallel line segment with respect to $\overline{AB}$ through point D ; draw a parallel line segment with respect to $\overline{AC}$ through B , such that two line segments intersect at point E and that $\overline{BC}$ intersect with $\overline{DE}$ at point F . Draw $\overline{CE}$. Prove that quadrilateral $BECD$ is a rectangle.



L10 E

4 With respect to quadrilateral $ABCD$, six conditions were given:

① $\overline{AB} \parallel \overline{CD}$; ② $\overline{BC} \parallel \overline{AD}$;

③ $AB = CD$; ④ $BC = AD$;

⑤ $m\angle A = m\angle C$; ⑥ $m\angle B = m\angle D$.

The number of ways to prove that $ABCD$ is a parallelogram by choosing two of the above conditions is _____ .

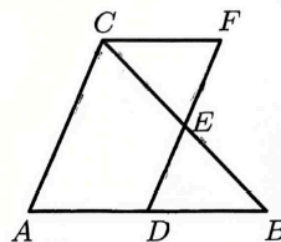
A. 4

B. 9

C. 13

D. 15

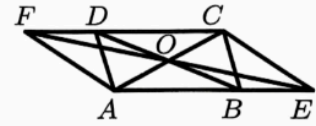
- 5 As shown in the figure, in $\triangle ABC$, D and E are the midpoints of $\overline{AB}$ and $\overline{BC}$, respectively, and the point F is on the extension of $\overline{DE}$. If a condition is added to make the quadrilateral $ADFC$ a parallelogram, then this condition is _____.



- A. $m\angle B = m\angle F$ B. $m\angle B = m\angle BCF$ C. $AC = CF$

- D. $AD = CF$

- 6 As shown in the figure, the diagonals of the parallelogram $ABCD$ intersect at point O , and the line EF passes through the point O , intersecting with the extensions of $\overline{AB}$ and $\overline{CD}$ at the point E and F , respectively. Prove: The quadrilateral $AECF$ is a parallelogram.



- As shown in the figure, draw a line EF to meet $\overline{AD}$ and $\overline{BC}$ at E and F , respectively through O , the intersection point of the diagonals of parallelogram $ABCD$. It is given that G and H are the midpoints of $\overline{OD}$ and $\overline{OB}$ respectively. Prove that quadrilateral $EHFG$ is a parallelogram.

